

The Hydrogen Atom

The Hydrogen problem

Time independent Schrödinger equ:

$$E \psi(\underline{x}) = \left(-\frac{\hbar^2}{2m} \Delta - \underbrace{\frac{d}{|\underline{x}|}}_{V(r)} \right) \cdot \psi(\underline{x})$$

$$d = \frac{e^2}{4\pi\epsilon_0}$$

$$(*) \quad \Delta \phi + \frac{2m}{\hbar^2} (E - \underbrace{V(r)}_{\text{any } V(r)}) \phi = 0$$

Solution in spherical coordinates:

$$\underline{x} = \begin{pmatrix} r \cdot \sin(\vartheta) \cdot \cos(\varphi) \\ r \cdot \sin(\vartheta) \cdot \sin(\varphi) \\ r \cdot \cos(\vartheta) \end{pmatrix}$$

Laplace operator in spherical coordinates:

$$\Delta = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} + \frac{1}{r^2 \sin^2 \vartheta} \frac{\partial^2}{\partial \varphi^2} + \frac{1}{r^2} \frac{1}{\sin \vartheta} \frac{\partial}{\partial \vartheta} \sin \vartheta \frac{\partial}{\partial \vartheta}$$

use separation of variables

$$\text{Ansatz} \quad \phi(r, \vartheta, \varphi) = R(r) \cdot \gamma(\vartheta, \varphi)$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial \phi}{\partial r} = \frac{\gamma}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right)$$

$$\frac{1}{r^2} \frac{1}{\sin^2 \vartheta} \frac{\partial^2}{\partial \varphi^2} \phi = \frac{R}{r^2} \cdot \frac{1}{\sin^2 \vartheta} \frac{\partial^2 \gamma}{\partial \varphi^2}$$

$$\frac{1}{r^2} \frac{1}{\sin \vartheta} \frac{d}{d\vartheta} \left(\sin \vartheta \frac{d\gamma}{d\vartheta} \right) \cdot R$$

substitute into (*), multiply by $\frac{r^2}{R \cdot \gamma}$

$\Rightarrow$

$$\frac{1}{\gamma} \left(\frac{1}{\sin^2 \vartheta} \frac{\partial^2 \gamma}{\partial \varphi^2} + \frac{1}{\sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial \gamma}{\partial \vartheta} \right) \right)$$

↖ this depends on ϑ, φ

$$+ \underbrace{\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\hbar^2}{2\mu} r^2 (E - V(r))}_{\text{const} = \ell(\ell+1)} = 0$$

↖ this line on r , only

$\Rightarrow$ they must be constants

second line yields: Radial Schrödinger eqn.

$$(1) \quad E R(r) = - \frac{\hbar^2}{2\mu} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left(V(r) + \frac{\hbar^2 \ell(\ell+1)}{2\mu \cdot r^2} \right) R(r)$$

$V_{\text{eff}}(r)$ effective potential

, describes radial motion of e^-

first line:

$$\frac{1}{\gamma} \left(\frac{1}{\sin^2 \vartheta} \frac{\partial^2 \gamma}{\partial \varphi^2} + \frac{1}{\sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial \gamma}{\partial \vartheta} \right) \right) = -\ell(\ell+1)$$

↓

angular Schrödinger eqn.

$$(2) \quad \frac{1}{\sin^2 \vartheta} \frac{\partial^2 \gamma}{\partial \varphi^2} + \frac{1}{\sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial \gamma}{\partial \vartheta} \right) + \ell(\ell+1) \gamma = 0$$

describes rotation of e^- around nucleus

• use separation of variables again in (2),

assume $\gamma(\vartheta, \varphi) = P(\vartheta) \cdot f(\varphi)$

• substitute into (2), multiply by $\frac{\sin^2 \vartheta}{P(\vartheta) \cdot f(\varphi)}$

$$\Rightarrow \underbrace{\frac{1}{f} \frac{d^2 f}{d\varphi^2}}_{\text{depends on } \varphi} + \underbrace{\frac{1}{P(\vartheta)} \sin \vartheta \frac{d}{d\vartheta} \left(\sin \vartheta \frac{dP}{d\vartheta} \right) + \ell(\ell+1) \sin^2 \vartheta}_{\text{depends on } \vartheta \text{ only}} = 0$$

$\Rightarrow$ must both be constant, and opposite in sign!

$$(2a) \quad \frac{1}{f} \frac{d^2 f}{d\varphi^2} \equiv -m^2 \quad (\text{positive does not work...})$$

$$(2b) \quad \frac{1}{P(\vartheta)} \sin \vartheta \frac{d}{d\vartheta} \left(\sin \vartheta \frac{dP}{d\vartheta} \right) + \ell(\ell+1) \sin^2 \vartheta = +m^2$$

(2a) can be solved easily:

$$\frac{d^2 f}{d\varphi^2} = -m^2 f \quad \Rightarrow \quad f(\varphi) = e^{im\varphi}$$

however $\phi(r, \vartheta, \varphi)$ must be continuous and single valued! Since φ and $\varphi + 2\pi$ describe the same point, therefore

$$\phi(r, \vartheta, \varphi + 2\pi) \stackrel{!}{=} \phi(r, \vartheta, \varphi)$$

$\rightarrow$ which is only possible if $f(\varphi + 2\pi) = f(\varphi)$,

$\rightarrow$ which is only satisfied if $m \in \mathbb{Z}$

m : turns out to be the magnetic quantum number, and will turn out to be related to $L_z \rightarrow L_z = \hbar m$

P then satisfies:

$$(3) \quad \frac{1}{\sin \vartheta} \frac{d}{d\vartheta} \left(\sin \vartheta \frac{dP}{d\vartheta} \right) + \left(\ell(\ell+1) - \frac{m^2}{\sin^2 \vartheta} \right) P(\vartheta) = 0$$

$\rightarrow$ solutions depend on $\ell, m \Rightarrow P_\ell^m(\cos(\vartheta))$

$$P_e^0 = P_e(\cos\theta)$$

Legendre polynomials

$$P_e^m$$

associated Legendre polynomials

l turns out to characterize the angular momentum, and to have proper solutions in (1) for $R(r)$, it must take values

$$l = 0, 1, 2, \dots$$

in this case m can take values in (3)

$$m = -l, -l+1, \dots, l-1, l \quad \text{or} \quad |m| \leq l$$

angular wave function is then

$$Y_l^m(\theta, \varphi) \sim P_l^m(\cos\theta) \cdot e^{im\varphi}$$

solutions of (1) shall be characterized by l and a quantum # n ($\leftrightarrow E$)

$$R(r) = R_{nl}(r)$$

yielding a solution:

$$\phi_{nlm}(r, \theta, \varphi) = R_{nl}(r) \cdot Y_l^m(\theta, \varphi)$$

↑
↑

main
quantum
nr.

angular momentum
 $l=0 \rightarrow s$
 $l=1 \rightarrow p$
 $\vdots$

magnetic

solution of (3)

$$x \equiv \cos \theta$$

$$\frac{d}{d \cos \theta} = - \frac{1}{\sin \theta} \frac{d}{d \theta}$$

$$\Rightarrow \frac{d}{dx} \left((1-x^2) \frac{df}{dx} \right) + \left(\ell(\ell+1) - \frac{m^2}{1-x^2} \right) f(x) = 0$$

$$\text{or } (1-x^2) f'' - 2x f' + \left(\ell(\ell+1) - \frac{m^2}{1-x^2} \right) f = 0$$

$$m=0 \Rightarrow P_\ell(x) \text{ Legendre polynomials}$$

$$P_\ell \sim \frac{d^\ell}{dx^\ell} \left((x^2-1)^\ell \right) \quad \ell = 0, 1, 2, \dots !$$

$\sim \ell$ 'th order polynomial of $\cos(\theta)$

$$m \neq 0 \text{ solution } P_\ell^m(x)$$

associated Legendre polynomial

$$P_\ell^m \sim (1-x^2)^{m/2} \frac{d^m}{dx^m} P_\ell$$

$$\bullet P_\ell^m \sim (\sin \theta)^m \times ((\ell-m) \text{'th order polynomial of } \cos \theta)$$

$$\bullet \frac{d^m}{dx^m} P_\ell = 0 \quad \text{if } m > \ell$$

Few examples :

$$P_0 = 1$$

$$\Rightarrow Y_0^0 \sim \text{cst}$$

$$P_1^0 \sim \frac{d}{dx} (x^2-1) \sim 2x \sim \cos \theta$$

$$\Rightarrow Y_1^0 \sim \cos \theta$$

$$P_1^1 \sim \sin \theta$$

$$\Rightarrow Y_1^{\pm 1} \sim \sin \theta e^{\pm i \cdot \varphi}$$

normalization must be worked out ...

Normalization:

$$1 \equiv \int d^3r |\phi(r)|^2 = \int_0^\infty dr \int_0^\pi d\vartheta \int_{-\pi}^\pi d\varphi \sin\vartheta \cdot r^2 |R(r)|^2 \cdot |Y(\vartheta, \varphi)|^2$$
$$= \left(\int_0^\infty dr \cdot r^2 |R(r)|^2 \right) \cdot \left(\int_{-\pi}^\pi d\varphi \int_0^\pi d\vartheta \sin\vartheta |Y(\vartheta, \varphi)|^2 \right)$$

$\uparrow \qquad \nearrow$
we set both to 1

$$\Rightarrow \iiint |Y_\ell^m(\vartheta, \varphi)|^2 \sin\vartheta d\vartheta d\varphi = 1$$

$$\int_{-\pi}^\pi d\varphi \int_0^\pi d\vartheta \sin\vartheta (Y_\ell^m(\vartheta, \varphi))^* \cdot Y_{\ell'}^{m'}(\vartheta, \varphi) = \delta_{\ell, \ell'} \delta_{m, m'}$$

orthonormality!
turn out to be eigenstates of $\hat{L}^2$ and $\hat{L}_z$!

$$\int_0^\infty dr \cdot r^2 |R(r)|^2 = 1$$

Y_ℓ^m are referred to as spherical harmonics:

"s"

$$Y_0^0 = \frac{1}{\sqrt{4\pi}},$$

"p"

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos\vartheta, \quad Y_1^{\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin\vartheta e^{\pm i\varphi}$$

"d"

$$Y_2^0 = \sqrt{\frac{5}{16\pi}} \cdot (3 \cos^2\vartheta - 1), \quad Y_2^{\pm 1} = \mp \sqrt{\frac{15}{8\pi}} \sin\vartheta \cos\vartheta e^{\pm i\varphi}$$

$$Y_2^{\pm 2} = \sqrt{\frac{7}{16\pi}} \sin^2\vartheta e^{\pm 2i\varphi}$$

Radial part

$$(1) \quad \frac{\hbar^2}{2\mu} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[E - \left(\frac{-e^2}{4\pi\epsilon_0 r} + \frac{\hbar^2}{2\mu} \frac{\ell(\ell+1)}{r^2} \right) \right] R = 0$$

we make it dimensionless (atomic units)

$$\rho \equiv \frac{r}{a_0} \quad a_0 \text{ Bohr radius} \quad a_0 = \frac{4\pi\epsilon_0 \hbar^2}{\mu \cdot e^2}$$

$$\text{first term, e.g.} \Rightarrow \frac{1}{2} \frac{\hbar^2}{\mu \cdot a_0^2} \left(\frac{d^2 R}{d\rho^2} + \frac{2}{\rho} \frac{dR}{d\rho} \right)$$

1 Hartree = 2 Rydberg $\approx 27.2 \text{ eV}$

Then (1) reads $\frac{E \cdot \mu a_0^2}{\hbar^2}$ measured in Hartrees

$$\frac{1}{2} \left(\frac{d^2 R}{d\rho^2} + \frac{2}{\rho} \frac{dR}{d\rho} \right) + \left(E + \frac{1}{\rho} - \frac{\ell(\ell+1)}{2 \cdot \rho^2} \right) R = 0$$

consider $\ell = 0$ case

$\ell = 0$

$$(1') \quad \frac{d^2 R}{d\rho^2} + \frac{2}{\rho} \frac{dR}{d\rho} + 2E R + \frac{2}{\rho} R = 0$$

guess: $R = e^{d \cdot \rho}$

this is a solution if $d = -1$ and if

$$d^2 = -2E \Rightarrow E = -\frac{1}{2} \Rightarrow E = -\frac{1}{2} \frac{\hbar^2}{\mu a_0^2}$$

$\Rightarrow R_0 \propto e^{-r/a_0}$ ← ground state!

$$E_0 = -\frac{1}{2} \frac{\hbar^2}{\mu a_0^2} = -\frac{1}{2} \frac{\mu \cdot e^4}{(4\pi\epsilon_0)^2 \cdot \hbar^2} = -1 \text{ Ry}$$

Normalization: $\int_0^\infty |R(r)|^2 r^2 dr = 1 \Rightarrow R_0 = \frac{2}{a_0^{3/2}} \cdot e^{-r/a_0}$

$\Rightarrow$ ground state wave function

$$\psi_{000}(r, \vartheta, \varphi) = \frac{2}{a_0^{3/2}} \cdot e^{-r/a_0} \cdot \frac{1}{\sqrt{4\pi}}$$

$\nearrow$ $\uparrow$ $\nwarrow$
 n l m

- no angular dependence ($\sim l=0$ "s" states)
- localized to region $r \propto a_0$

$l=0$ excited states ?

notice $\frac{d^2 R}{d\rho^2} + \frac{2}{\rho} \frac{dR}{d\rho} = \frac{1}{\rho} \frac{d^2}{d\rho^2} (\rho R)$

$\underbrace{}_{u = \rho \cdot R(\rho)}$

u' can be written as

$$\frac{1}{2} \frac{d^2 u}{d\rho^2} + \left(\epsilon + \frac{1}{\rho} \right) u = 0$$

for ground state $u = \rho e^{-\rho}$

try $u(\rho) = Q_n(\rho) e^{-\rho}$

$\uparrow$ order n polynomial (Laguerre)

$$Q_n = a_1 \rho + \dots + a_n \rho^n$$

• leading terms

$$\rho \rightarrow \infty \Rightarrow \sim \rho^n \cdot e^{-\rho} \Rightarrow -\frac{d^2}{2} = \epsilon$$

coefficients of $\sim \rho^{s-1}$ term $\rightarrow$ recursion relation

$$\frac{1}{2} \left(\cancel{d^2 a_{s-1}} - 2 \cdot 2s a_s + s(s+1) a_{s+1} \right)$$

$$+ \epsilon \cdot \cancel{a_{s-1}} + a_s = 0$$

$-\frac{d^2}{2}$

$$\Rightarrow a_{s+1} = \frac{2 \cdot 2s - 2}{s(s+1)} \cdot a_s$$

since $a_{n+1} = 0 \Leftrightarrow 2n = 1 \Rightarrow d = \frac{1}{n} !$

$$\Rightarrow E = - \frac{1}{2 \cdot n^2}$$

$$\Rightarrow E_n = - \frac{1}{2 \cdot n^2} \frac{\hbar^2}{\mu a_0^2} = - \frac{1}{n^2} R_y \text{ Rydberg}$$

$\Rightarrow$ solution ($l=0$)

$$u(\rho) \sim Q_n^1(\rho) e^{-\rho/n}$$

$$E_n = - \frac{\hbar^2}{2} \frac{1}{\mu a_0^2} =$$

$$\Rightarrow R_{n0}(r) \sim \frac{1}{r} Q_n^1\left(\frac{r}{a_0}\right) e^{-\frac{r}{n \cdot a_0}}$$

$\uparrow$
 e

$\nearrow$
 "radius" $\sim n \cdot a_0$!

$$E_n = - \frac{1}{n^2} \frac{\hbar^2}{2 \mu a_0^2}$$

$l \neq 0$ excited states

$$\frac{1}{2} \frac{d^2 u}{d \rho^2} + \left(E + \frac{1}{\rho} - \frac{l(l+1)}{2 \cdot \rho^2} \right) u(\rho) = 0$$

$\nearrow$ most singular at $\rho \rightarrow 0$

$\Rightarrow$ try

$$u = (a_k \rho^k + \dots + a_n \rho^n) e^{-\alpha \cdot \rho}$$

again $E = - \frac{\alpha^2}{2} \quad (\rho \rightarrow \infty)$

but $\rho \rightarrow 0 \Rightarrow \frac{1}{2} a_k k(k-1) \rho^{k-2} = \frac{1}{2} l(l+1) a_k \rho^{k-2}$

$$\Rightarrow k = l+1$$

$$u(\rho) = \underbrace{(a_{\ell+1} \rho^{\ell+1} + \dots + a_n \rho^n)}_{Q_{\ell+1}^n(\rho)} e^{-\alpha \cdot \rho}$$

recursion yields again coefficients and

$$\alpha = \frac{1}{n} !$$

Full solution

$$\phi_{n\ell m}(r, \vartheta, \varphi) \sim R_{n\ell}(r) Y_{\ell m}(\vartheta, \varphi)$$

$$R_{n\ell}(r) \sim \frac{1}{r} Q_{\ell+1}^n\left(\frac{r}{a_0}\right) e^{-\frac{r}{n \cdot a_0}}$$

$$Y_{\ell m}(\vartheta, \varphi) \sim P_{\ell}^{|m|}(\vartheta) e^{i m \cdot \varphi}$$

Quantum numbers :

- $n = 1, 2, \dots$
- $\ell = 0, 1, \dots, \ell < n$
- $m = -\ell, \dots, \ell$

$$E_n = -\frac{1}{n^2} E_1$$

$$E_1 = 1 \text{ Ry} = 13,6 \text{ eV}$$

Few examples :

$$n=1 \quad \rightarrow \quad R_{10} = 2 a^{-3/2} e^{-r/a_0}$$

$$n=2 \quad \rightarrow \quad R_{20} = \frac{1}{\sqrt{2}} a^{-3/2} \left(1 - \frac{1}{2} \frac{r}{a_0}\right) e^{-r/2a_0}$$

$$\rho \quad R_{21} = \frac{1}{\sqrt{24}} a^{-3/2} \frac{r}{a_0} e^{-r/2a_0}$$

$$n=3 \quad \rightarrow \quad R_{30} = \frac{2}{\sqrt{27}} a^{-3/2} \left(1 - \frac{2}{3} \frac{r}{a_0} + \frac{2}{27} \left(\frac{r}{a_0}\right)^2\right) e^{-r/3a_0}$$

⋮